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When to Replan and How Much to Correct: Residual Reinforcement Learning for Disturbance Recovery in Vision-Language-Action Policies

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***Abstract*—Vision-language-action policies execute predicted action chunks open loop, so when an object is suddenly displaced, the robot keeps following a plan made for the old scene. On LIBERO-10, teleporting the target object by 4 to 6 cm, mostly as the gripper approaches, cuts the pi0.5 policy's success from 95 to 65 percent. Under one paired protocol, simple inference-time levers (replan frequency, integration steps, chunk selection) recover little, and replanning recovers nothing measurable at the next scheduled query but about three points, at unchanged inference cost, when a learned camera-frame detector triggers it at the event. Correcting the actions recovers more: a fixed-rule shift toward the estimated displacement gains about six points, and an event-gated residual, trained with reinforcement learning on privileged state as lightweight post-training and distilled into a vision-only student, raises held-out success from 62.2 to 71.8 percent (gain 9.6 points, 95 percent interval 5.8 to 13.4; on all 500 episodes 11.2, and 8.3 with displacements twice as large) and keeps undisturbed success at 95.4 percent. The detector, trained on teleports, rarely fires when an object slides or the arm is pushed.**

***Index Terms*—vision-language-action models, post-training, residual reinforcement learning, action chunking, event-triggered replanning, disturbance detection**

I. Introduction

Vision-language-action (VLA) models have become a standard recipe for language-conditioned manipulation [1]–[3]. Most now predict *action chunks* [4], [5]: one query of the model yields a short sequence of H actions, of which the robot executes a prefix of h steps before the model is queried again. Chunking smooths motion and amortizes the backbone's cost but makes the controller open loop between queries. If the scene changes inside a chunk, the robot keeps executing actions planned for the old scene until the chunk is spent, and the next query sees a configuration that the training data rarely contains.

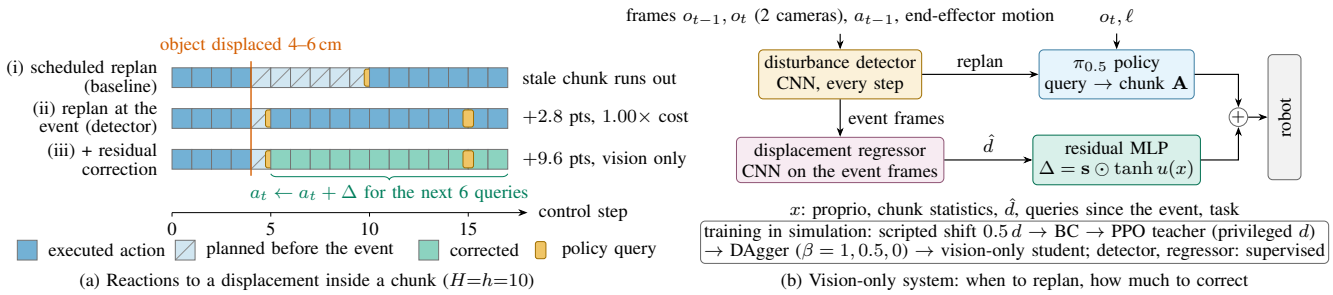
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How large is this cost, and what recovers it? We study the question with the official $\pi_{0.5}$ checkpoint for LIBERO [3], [6] under a controlled disturbance: when the open gripper first comes within 12 cm of the object it is approaching (or at a random step if it has not by then), the object is teleported 4–6 cm horizontally, which cuts success on LIBERO-10 from 95.4% to 64.7%. Tracking the object shows what goes wrong (Section III): in about six of ten episodes, successful or not, the policy first closes the gripper nearer the *old* position than the new one, and then either recovers within a few attempts or loops until the episode times out.

Recent work offers several responses. Adaptive-inference methods decide when to replan, how much to compute per query, or which of several sampled chunks to keep [7]–[10]; runtime monitors detect execution failures [11]–[13]; and residual reinforcement learning (RL) learns corrections on top of a fixed policy [14], [15]. They are evaluated on different tasks and disturbances, so it is unclear how much each kind of mechanism recovers when a static scene changes abruptly; we test simple versions of each kind under one protocol. The inference levers recover little (Section V-A). Replanning *at* the displacement helps where waiting for the next scheduled query does not, but by only about three points, which a learned detector realizes at unchanged inference cost; the new chunk seems to correct only partly toward the new position, so the actions may need correcting. A fixed-rule shift toward the estimated displacement recovers about six points. A residual policy trained with RL as lightweight post-training around the VLA, gated by the detector and distilled into a student without privileged inputs, recovers about ten of the 31 lost points (9.6 on held-out initial states, 11.2 on all 500 episodes; Fig. 1).

Our contributions are:

- a controlled mid-execution displacement protocol for LIBERO-10 (complementing studies that perturb the setup before execution [16] or the robot's actions during it [17], [18]), a diagnosis of how a chunked VLA fails, and an evaluation protocol that compares every variant with the mean of seven sampling-noise draws of the base policy on identical episodes;



(a) Reactions to a displacement inside a chunk (H=h=10)

(b) Vision-only system: when to replan, how much to correct

Fig. 1. Overview. (a) A chunked policy that is queried every h actions cannot react to a displacement until the current chunk is spent (i). A learned detector interrupts the chunk and replans at the event (ii), and a residual policy adds a small correction Δ to every action of the following queries (iii). (b) All three learned modules use only camera frames, actions, proprioception and the task; the $\pi_{0.5}$ weights are not updated. Gains: held-out episodes, as in Fig. 2.

- a side-by-side comparison showing that simple inference levers do not recover the loss and that plain replanning recovers about three of the 31 points when timed at the event and nothing measurable at the next scheduled query, with a learned per-step detector that recognizes 87–90% of held-out displacements within three control steps;
- an event-gated residual recovery policy, trained as a privileged teacher by behavior cloning (BC) of a scripted shift followed by proximal policy optimization (PPO) and distilled into a vision-only student, that adds 3.9 points over a fixed-rule shift on held-out initial states (4.7 on all 500), transfers to displacements twice as large and does not harm undisturbed episodes.[1]

II. Related Work

Adaptive chunk execution. AAC sets the executed prefix of a chunk from the entropy of sampled chunks [7], bidirectional decoding selects among chunks resampled at every step [10], RTC inpaints the next chunk while the current one executes [19], DVAC executes the low-variance prefix of a chunk and replans before its high-variance actions [20], and DA-SIP chooses the number of integration steps per query with a difficulty classifier [9]. They balance reactivity, consistency and compute but detect no events. VLA-Corrector [8] does: it truncates the chunk when visual latents drift from a learned prediction and replans with gradient guidance, but adds no correction policy.

Runtime monitoring. Sentinel checks temporal action consistency and, with a vision-language model, task progress [11], FAIL-Detect applies conformal sequential out-of-distribution detection [12], and SAFE learns a VLA failure score from internal features [13]; these monitors flag failures or halt the policy. Like VLA-Corrector, we use detection to *act*, but on one specific external event, and we also gate a learned correction.

Correcting a fixed policy's actions. Residual RL [21] refines imitation-learned policies [14], [22] and a frozen π_0, with the corrections then distilled into the VLA [15], and latent-noise RL steers π_0 without changing its weights [23]. Other work post-trains the VLA itself with online RL [24], [25], which has been shown to help when an object is moved mid-episode [26], or with offline RL to recover from errors [18]. A training-free operator offsets and compresses a chunk toward perceived motion [27], the closest analogue of our fixed-rule shift. None of these corrections is gated by an event; ours is zero except after a detected displacement, distilled from a privileged teacher into a vision-only student as in [22], and compared with a fixed-rule shift on the same episodes.

[1]Code, trained modules and per-episode results: https://github.com/BitAdventurer/VLA-Disturbance-Recovery

III. Setting and Diagnosis

A. Chunked Execution and the Disturbance

A flow-based VLA maps an observation o_t and instruction ℓ to a chunk $\mathbf{A}_t = (a_t, \ldots, a_{t+H-1})$ by integrating a learned velocity field for n Euler steps [2], [28]; the robot executes $h \leq H$ actions and queries again. The same family of conditional generative models served our brain-network generation [29]. We use the released $\pi_{0.5}$ LIBERO checkpoint of openpi [3] with $H = 10$ and, unless stated otherwise, $(n, h) = (2, 10)$ (95.4% undisturbed); openpi's example setting $(10, 5)$ reaches 91.9% (three draws) at $2.8\times$ the cost and loses as much as $(2, 10)$ under the displacement defined below (to 61.2%, two draws). Episodes follow the released script (520 control steps at 20 Hz after 10 settling steps) on the 50 fixed initial states of each of the ten tasks, 500 episodes per run.

The *object displacement* is applied once per episode: when the open gripper first comes within 12 cm (end effector to object center) of a movable object of interest (an object named in the task goal) that it does not hold, that object is teleported horizontally by a distance drawn from $[4, 6]$ cm in a uniformly random direction and its velocity is zeroed. If no approach has occurred by a random step in 50–170 (about 26% of episodes, mostly in two tasks), the nearest free object of interest is displaced at that step. Direction and magnitude are seeded by the task and initial state, so every method sees identical displacement vectors. Success drops to 64.7% (mean of seven noise draws), and every failure is a timeout.

B. Why the Policy Fails

We tracked the displaced object in 100 disturbed episodes (initial states 0–9 of each task, one base draw; Table I). All

TABLE I
WHY EPISODES FAIL AFTER THE DISPLACEMENT (100 TRACKED EPISODES)

	Success (65)	Failure (35)
Displaced object eventually lifted	100%	54%
Gripper closures before the lift	2.3	7.7
First closure nearer to the old position	62%	63%
Motion toward old / new position[a]	0.69 / 0.60	0.26 / 0.39

[a]Mean cosine of end-effector motion with the direction to each position over the 15 steps after the event.

successes eventually lift it, but 46% of failures never do. In about 62% of *both* groups the first gripper closure after the event is nearer to the object's old position than to its new one, and even successful episodes at first move more toward the old position (cosine 0.69 versus 0.60); failures move decisively toward neither (0.26 / 0.39). The new position is visible at the next query, yet replanning then does not help (Section V-B), and recovery depends on whether a few regrasps succeed before time runs out. This suggests that replanning, even at the event, may not re-target the moved object and that the actions themselves may need to be shifted; Section V tests both.

C. Evaluation Protocol

Flow-based policies sample the initial noise of the integration, and this noise alone moves the success of $(2, 10)$ on the disturbed episodes between 63.2% and 67.8% across seven seeds ("noise draws"), so single draws are noisy. We compare every variant with the *per-episode mean of the seven noise draws* of the base policy on the same 500 (task, initial state) episodes with the same disturbance, and report the mean paired difference with a task-stratified bootstrap 95% interval (6,000 resamples). Variants run once carry that draw's noise and resolve only differences of several points. Detector, regressor and teachers are each trained once and the students with three distillation seeds (the released one unless stated), all on initial states 0–24 only; initial states 25–49 (250 episodes, base 62.2%) are held out from training for the main comparison (Fig. 2); results on all 500 episodes include the training initial states. States 25–49 also served to choose gates, α and model versions, so they are held out from training but not from model selection; intervals are per comparison, without multiplicity correction.

IV. METHOD

A. Detecting the Displacement and Replanning at Once

The detector runs beside the controller at every control step and receives, for each of the two cameras (third-person and wrist), the previous frame, the current frame and their difference at 64×64 grayscale, together with the action executed between them and the measured end-effector motion; a four-layer CNN and a small MLP fuse these into one logit (0.1 M parameters). Hand-crafted signals (camera, backbone-embedding and chunk changes) reach query-level AUROCs of only 0.50–0.83 on training-state episodes, whereas this network detects 87–90% of held-out displacements within three steps. Positives are the frame pairs straddling a displacement, hard negatives the pairs within 15 steps of it, plus 250 undisturbed training-state episodes; training on the 500 events of two disturbed runs that fall on the training initial states takes seven minutes, and the threshold is set on that split for 0.05 false alarms per undisturbed episode.

When the logit crosses the threshold, the schedule interrupts the current chunk and queries the policy immediately, so within a few control steps of the event the robot executes a chunk planned for the new scene (Fig. 1(a)). Inference cost, the policy-query time per episode from the measured per-query latency, stays at $1.00\times$ the base schedule: each detection (1.07 per disturbed episode) adds at most one query (1.7% more queries per step), but episodes end earlier on average; a variant that also spends $n = 10$ steps on the three queries after the event costs $1.02\times$.

B. Residual Correction

We add a residual $\Delta \in \mathbb{R}^3$ to the translational part of every action of the chunk returned by a query,

$$a_j^{\text{xyz}} \leftarrow \text{clip}(a_j^{\text{xyz}} + \Delta, -1, 1), \qquad \Delta = \mathbf{s} \odot \tanh u(x), \quad (1)$$

with $\mathbf{s} = (0.3, 0.3, 0.15)$ in LIBERO action units (0.3 units move the end effector by about 3.6 mm per control step), computed once per query from a feature vector x by a two-layer MLP u with 256 units. The residual is *gated*: it is zero except for a bounded window after a detected displacement, so undisturbed behavior is touched only on false alarms.

Scripted shift (privileged). As a probe, once the simulator reports at a scheduled query that an object of interest has moved by d since the reset (more than 2 cm, not lifted), we add αd divided equally over the kH actions of the next k queries ($0.5\,d$ is about 0.1 units per action), horizontal only and without interrupting the chunk. With $k = 2$, $\alpha = 0.5$ and 1 perform alike (72.2% and 71.2% on all 500 episodes, one draw), while $\alpha = 1.5$ falls to 64.8%, presumably because overcorrecting adds to the base policy's own partial correction. A vision-only counterpart, the *fixed-rule shift*, replans at each detection and adds $0.5\,\hat{d}$ in the same way, with $\hat{d}$ from the displacement regressor below.

Privileged teacher. We then learn the correction with residual RL [21] on privileged state. Deciding how much to intervene at every gated query is a sequential decision problem of the kind we have solved with RL for brain-connectivity segmentation [30], graph sparsification [31] and agentic diagnosis [32]. The teacher's 53 inputs are proprioception, a summary of the base chunk, the relative position, displacement and presence of up to three objects of interest, episode progress, gripper state, queries since the gate opened, the previous residual and a task one-hot. Its gate, checked at every scheduled query (no replan at the event), opens when an object has jumped more than 2.5 cm since the previous query while resting at its initial height (within 1.5 cm), with the gripper open and more than 5 cm away, and closes once the object is lifted 3 cm. It opens in 85% of disturbed and in none of 250 undisturbed

held-out episodes; a cumulative 2-cm gate opened in 54% of the undisturbed ones and, with a learned teacher behind it, cut their success to 73.6%. The actor is Gaussian over u (initial standard deviation 0.30). We first clone the scripted shift ($\alpha = 0.5$, $k = 2$) from about 450 episodes, then run PPO with generalized advantage estimation [33] (settings in the code release) for 40 iterations of 120 episodes on the training initial states. The reward is the episode success minus $0.01\,\|\Delta/\mathrm{s}\|^2$ per gated query, a penalty whose weight we fixed rather than adapted as in our earlier RL pipelines [34], [35]; one run takes about two GPU-hours.

Vision-only student. The student may use neither object positions nor the privileged gate. Its gate is the learned detector, active for six queries after each detection; a firing inside the window restarts it, and every firing, true or false, costs one replan. When the detector fires, a *displacement regressor* estimates the horizontal displacement $\hat{d}$ from the 128×128 RGB frame pair of both cameras at the firing step (four ResNet blocks, 0.5 M parameters; 1,499 training-state events of 2–10 cm, 1,499 held-out events: median error 1.6 cm at a median displacement of 6.1 cm, median direction error 9°). The student MLP sees proprioception, the base-chunk summary, $\hat{d}$ and its norm, an active flag, the queries since the gate opened, episode progress, the gripper state, the previous residual and the task one-hot (36 inputs), so it can weigh the explicit estimate against the implicit drift of the base chunk, in the spirit of our adaptive fusion of static and dynamic connectivity features [36]. It is distilled from the teacher's mean action with DAgger [37]: three rounds of two rollouts per training initial state under the student's gate and replan, the teacher's residual replacing the student's at each gated query with probability $\beta = 1, 0.5, 0$. Every gated query is labeled with the teacher's residual (zero where its privileged gate is closed; 8,148 labels) and the student is refit by least squares on Δ.

V. Results

A. Inference Levers Recover Little

Fig. 2 (top, held out) shows simple versions of the levers that adaptive-inference methods tune. Replanning twice as often, $(2, 5)$, trends down (-4.2) and five times as often, $(2, 2)$, loses ten points (87.9% undisturbed), and every schedule tested falls 29–33 points below its own undisturbed success (all 500). Five times more integration steps, $(10, 10)$, change nothing measurable ($+1.0$). Selecting one of $K = 8$ sampled chunks for the seven queries after the (oracle-timed) event also recovers little: random -0.6, medoid $+0.6$, a privileged rule aiming at the displaced object $+3.0$. Run once, $(10, 10)$ and chunk selection cannot exclude small gains.

B. When to Replan: At the Event

On held-out episodes, replanning only from the *next scheduled* query gains nothing measurable even with $n = 10$ and oracle event timing ($+0.6$ [-2.6, $+4.1$]). Replanning *at* the displacement with the same oracle gains $+3.7$ at $(2, 10)$ and $+4.5$ with $n = 10$ for the following queries, the latter

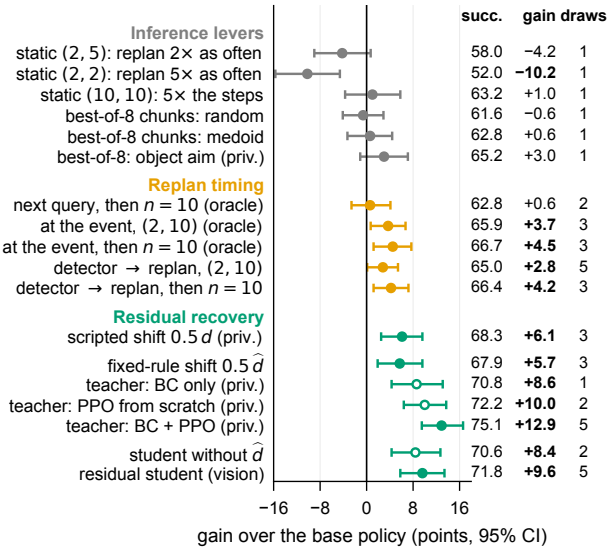


Fig. 2. Gain on the 250 held-out disturbed episodes over the base policy $(2, 10)$ (mean of seven noise draws, 62.2%); task-stratified paired bootstrap 95% intervals. (n, h): integration steps and executed actions per query; "oracle": event time from the simulator; "priv.": uses simulator state; "static": fixed schedule; hollow: ablations. Right: success, gain (bold when the interval excludes zero), noise draws.

$+3.9$ [$+0.4$, $+7.3$] above the next-query replan. The learned detector realizes most of this gain without the oracle ($+2.8$ and $+4.2$; all 500: Table II), at $1.00\times$ and $1.02\times$ the inference cost. In closed loop it detects 87–90% of the held-out displacements within three control steps (98–100% on the training initial states), with 0.10–0.15 false alarms per disturbed episode, and leaves undisturbed success at 95.2% on all 500 (base 95.4%). Even at the event, a replan recovers only a small part of the loss (Section V-C).

C. How Much to Correct: Residual Recovery

On held-out episodes the privileged scripted shift and the fixed-rule shift by $0.5\,\hat{d}$, which adds a replan at each detection, gain about the same ($+6.1$ and $+5.7$, three draws each; Fig. 2). With privileged d, the detector's replan plus the scripted shift reaches 71.3% held out (three draws), $+6.4$ [$+2.9$, $+10.1$] above the replan alone: shifting the actions adds to replanning. Learning the correction helps further: the privileged teacher, whose correction is longer and more gradual than the scripted shift's, reaches 75.1% on held-out episodes ($+12.9$ [$+9.5$, $+16.6$], five draws), $+4.3$ [$+0.5$, $+8.4$] above its behavior-cloned start (one draw). PPO from scratch reaches 72.2% (two draws), 2.9 points [-0.4, $+6.3$] below the teacher, so RL alone recovers most of the gain. With the detector's replan the teacher reaches 76.1% ($+1.0$, three draws), so on top of a learned correction the replan's share is not resolved.

The distilled student comes within a few points of its teacher without any privileged input: 71.8% held out ($+9.6$ [$+5.8$, $+13.4$], five draws; -4.4 [-7.3, -1.6] against the teacher with the same replan) and 75.9% on all 500 episodes (Table II; 75.9–76.8% over three distillation seeds). Held out, it is $+3.9$

TABLE II
RECOVERY ON 500 DISTURBED AND ON UNDISTURBED EPISODES
(BOLD: VISION-ONLY STUDENT)

Method	In.	Draws	Succ.	Gain [95% interval]	Undist.
$\pi_{0.5}$ alone, $(2, 10)$	–	7	64.7	–	95.4
Detector → replan, $(2, 10)$	V	5	67.2	$+2.5$ $[+0.6, +4.5]$	95.2
Detector → replan, then $n{=}10$	V	3	67.9	$+3.3$ $[+1.0, +5.5]$	–
Scripted shift $0.5\,d$	P	3	70.9	$+6.3$ $[+3.9, +8.7]$	90.8
Fixed-rule shift, $0.5\,\hat{d}$	V	3	71.2	$+6.5$ $[+3.7, +9.3]$	95.4
Residual teacher (BC + PPO)	P	5	77.4	$+12.7$ $[+10.1, +15.3]$	95.2
Student without $\hat{d}$	V	2	74.5	$+9.8$ $[+6.9, +12.7]$	95.4
Residual student	V	5	**75.9**	$\mathbf{+11.2}$ $[+8.5, +13.9]$	95.4

In.: V vision + proprioception, P privileged state; Undist.: undisturbed, 1 draw.
Succ.: mean over draws; half of the 500 episodes use training initial states.

$[+0.9, +6.9]$ above the fixed-rule shift and $+3.9$ $[+1.2, +6.8]$ above the rule selected on training initial states among seven settings ($\alpha = 0.5$ over $k = 4$ queries, three draws; all 500: $+4.7$ and $+4.8$). On undisturbed episodes the student (active in 5% through false alarms) scores 95.4%, as does the base policy; the pipeline adds a sub-millisecond CNN per step, one regressor pass per firing and one MLP per query to a 33 ms policy query (one RTX 5090).

D. Generalization and Side Effects

The student and detector were trained only on 4–6 cm displacements, mostly during the approach (the regressor on 2–10 cm); Fig. 3 evaluates them unchanged on other disturbances (all 500 episodes). With 8–12 cm displacements, where the base policy falls to 37.0% (three draws), the student gains $+8.3$ $[+5.0, +11.5]$ and the privileged teacher $+12.1$ (two draws each; $+5.8$ to $+8.8$ over three student seeds), while the fixed-rule shift gains only $+3.8$ (three draws). The student is $+4.5$ $[+1.5, +7.5]$ above it ($+4.2$ above the selected rule, two draws), so its lead over the fixed rule persists. On the other disturbances the student is neutral: for a late move of the nearest free object (steps 100–200) the detector fires in half of the episodes and the student's gain is within noise ($+1.4$), whereas the privileged teacher is slightly harmful (-2.4 $[-4.2, -0.5]$, two draws each); for sliding (6–10 cm planned, mostly cut short) and a five-step arm push the detector, trained on teleports, rarely fires (6–12%) and success changes by at most 1.1 points. Undisturbed success is 95.2–95.6% in every detector-gated variant measured: the detector gate bounds how often a residual can act, whereas the scripted shift's cumulative gate costs 4.6 points (90.8%).

E. Ablation: Displacement Estimate

A student distilled without $\hat{d}$ is only slightly worse than the full student on all 500 episodes (-1.4 $[-3.5, +0.9]$, two draws; -1.4 to -2.8 over three distillation seeds): at 4–6 cm the detector tells it *when*, while the base chunk, proprioception and the task input may tell it *how*. At 8–12 cm $\hat{d}$ helps in two of three seeds (without it: -3.9 $[-6.5, -1.3]$, two draws; other seeds -6.0 and $+0.8$). It does not make the correction proportional: held out, from 4–6 to 8–12 cm displacements the student's shift along the displacement grows only from 2.3 to 3.2 cm (the teacher's from 3.3 to 9.2 cm), as the regressor saturates near 6 cm.

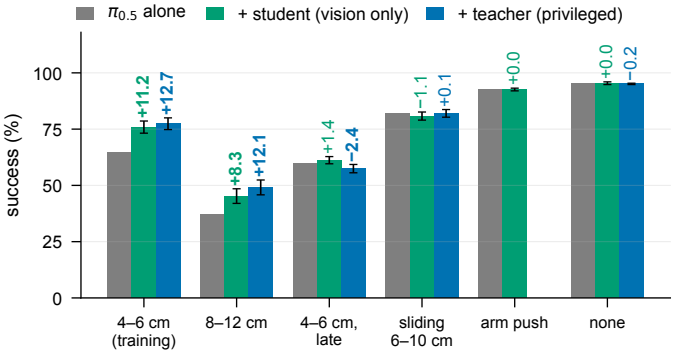


Fig. 3. Transfer from the training disturbance (left) to unseen ones and to "none" (500 episodes each; baselines: seven draws at 4–6 cm, three at 8–12 cm, two for late and sliding, one otherwise; methods: five draws at 4–6 cm, two at 8–12 cm and late, one otherwise; the teacher was not run with the arm push). Paired gains with 95% intervals, bold when excluding zero.

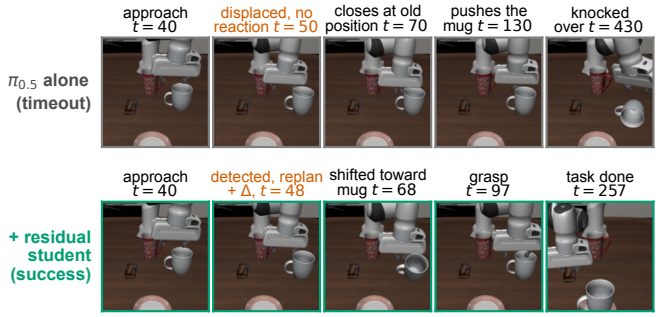


Fig. 4. One held-out episode of the mug-and-pudding task, same initial state and sampling noise; t is the control step. Top: the base policy alone (timeout). Bottom: with the detector and residual student (success).

F. Where the Residual Helps and Where It Fails

Gains concentrate where the approach triggers the displacement: held out, the student gains $+12.3$ $[+7.6, +17.1]$ on the 179 approach-triggered episodes but $+2.7$ $[-2.6, +7.9]$ on the 71 where the time fallback moves an object before the gripper nears it; there the privileged teacher still gains $+8.1$ and the detector's replan alone -1.8. On all 500, the student solves 36.5% of its runs on the 63 episodes the base policy never solves in seven draws and keeps 95.1% on the 199 always solved. The residual was active in 91% of the student's remaining failures (85% held out), so missed detections explain few of them; the correction itself, labeled zero where the teacher's gate stayed closed, is a likely bottleneck.

In the held-out episode of Fig. 4, the mug is displaced 5.1 cm at $t = 47$; the base policy closes the gripper where the mug used to be, pushes it around, knocks it over and times out. With the detector and student, a new chunk arrives one step after the event and the mug is grasped at $t = 97$. The regressor's estimate is poor here (1 cm, wrong direction), yet the correction points toward the mug, perhaps from the base chunk's drift or a task-level push.

VI. Conclusion

When the object it approaches is suddenly displaced, $\pi_{0.5}$ often first closes the gripper nearer its old position and loses 31 points. Inference-time levers recover little; replanning at the detected event recovers about three points, a fixed-rule shift about six, and an event-gated residual learned with RL ten (held out) to eleven (all 500). The gate keeps this cheap post-training conservative: a loosely gated teacher cut held-out undisturbed success from 95.2% to 73.6%; the gated residual leaves it unchanged. Limitations are one policy and benchmark, teleports, synchronous inference in simulation, most modules trained once and held-out states reused for model selection. Next, we will test whether the correction responds to the displacement or acts partly as a task-level push, compare it with fine-tuning the policy, and add physical disturbances, asynchronous execution, an end-to-end gate (possibly as cooperating agents [38]) and hardware.

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